Saarthi Manual

Section 4: Software Setup + Installation on Raspberry Pi 5

🎯 4.1 GOAL OF THIS SECTION

By the end of this section, you will have:
✅ Installed the right OS
✅ Connected to your Pi via monitor or SSH
✅ Installed Python libraries
✅ Set up the Google Coral USB Accelerator
✅ Verified your camera and sensors
✅ Prepared to run your first real-time AI model

🛠️ 4.2 TOOLS NEEDED

Tool	Purpose
Raspberry Pi Imager	To flash the OS onto SD card
32GB+ microSD card (Class 10)	Storage for OS + packages
HDMI monitor, keyboard, mouse	For initial setup (optional)
Wi-Fi internet	Required for installing packages
USB-to-USB-C cable	To power the Pi via power bank

🧠 4.3 STEP-BY-STEP OS INSTALLATION

🔹 Step 1: Flash Raspberry Pi OS

1. Download **Raspberry Pi Imager** from:
   https://www.raspberrypi.com/software
2. Insert microSD card via card reader.
3. Open Imager:
   - OS: Choose "Raspberry Pi OS (64-bit)"
   - Storage: Select your SD card
   - Click "⚙️" (advanced settings) and:
     - Enable SSH (if headless)
     - Set hostname = `saarthi-pi`
     - Set Wi-Fi name and password
     - Set username: `pi`, password: `raspberry` (or your choice)
4. Click **"WRITE"** and wait.

5. Insert SD into Pi, power it ON.
   Connect to monitor OR SSH using:

   ```
   ssh pi@saarthi-pi.local
   ```

🔧 4.4 INITIAL SYSTEM CONFIGURATION

```
sudo raspi-config
```

Go to:

- **Interface Options** → Enable:
  - Camera
  - I2C (for compass in future)
  - SSH
- **Localization** → Set locale to `en_IN.UTF-8`

Reboot:

```
sudo reboot
```

🪢 4.5 INSTALL PYTHON + REQUIRED LIBRARIES

Update your Pi:

```
sudo apt update && sudo apt upgrade -y
```

Install essentials:

```
sudo apt install python3-pip python3-opencv libatlas-base-dev libopenjp2-7 -y
pip3 install numpy RPi.GPIO imutils pyttsx3
```

Install camera tools:

```
sudo apt install libcamera-apps -y
```

□ 4.6 INSTALL CORAL USB ACCELERATOR SUPPORT

1. Enable USB Coral:

```
echo "deb https://packages.cloud.google.com/apt coral-edgetpu-stable main" | sudo tee
/etc/apt/sources.list.d/coral-edgetpu.list
curl https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo apt-key add -
sudo apt update
sudo apt install libedgetpu1-std
```

2. Test detection:

```
lsusb
```

```
# Look for "Google Inc." entry
```

📷 4.7 TEST CAMERA & SENSORS

Camera test:

```
libcamera-still -o test.jpg
```

Ultrasonic sensor test:
Reuse the Python script from Section 3.

🔊 4.8 TEST AUDIO OUTPUT

Install text-to-speech:

```
sudo apt install espeak
```

Test voice:

```
espeak "Saarthi is ready"
```

To adjust volume:

```
alsamixer
```

🔄 4.9 OPTIONAL: AUTO-START SCRIPT ON BOOT

You can auto-run your main AI script on startup:

```
sudo nano /etc/rc.local
# Add this line ABOVE `exit 0`:
python3 /home/pi/saarthi/main.py &
```

✅ 4.10 CHECKPOINT

By now, you should be able to:

- Boot into Pi OS
- Access camera and sensors
- Use Coral USB
- Hear speech output
- Install any Python libraries needed

□ FINAL NOTE:

You now have a fully working **software base** ready for AI logic and real-time alerts!

In the next section, you'll:

- Load the MobileNet SSD v2 `.tflite` model
- Run real-time camera inference
- Integrate object detection with audio feedback and sensors